Django ML Prediction API – Student Performance Index

This guide shows **end- to- end** how to:

1. Create a **Django project**
2. Store the dataset in the database
3. Expose the data via an **API**
4. **Train & deploy** a machine- learning model
5. **Predict** performance for new data entries

Dataset target: **Performance Index (Regression)**

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1. Project Setup

```
python -m venv venv
source venv/bin/activate  # Windows: venv\Scripts\activate

pip install django djangorestframework pandas scikit-learn joblib

django-admin startproject student_ml
cd student_ml
python manage.py startapp performance
```

Enable apps in `settings.py`:

```
INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    'rest_framework',
    'performance',
]
```

2. Database Model (`performance/models.py`)

```
from django.db import models

class StudentPerformance(models.Model):
    hours_studied = models.IntegerField()
    previous_scores = models.IntegerField()
    extracurricular = models.BooleanField()
    sleep_hours = models.IntegerField()
    sample_papers = models.IntegerField()
    performance_index = models.FloatField()

    def __str__(self):
        return f"Performance: {self.performance_index}"
```

```
python manage.py makemigrations
python manage.py migrate
```

3. Load Dataset into Database (One-time Script)

Create `performance/load_data.py`

```
import pandas as pd
from performance.models import StudentPerformance


def run():
    df = pd.read_csv('dataset.csv')

    for _, row in df.iterrows():
        StudentPerformance.objects.create(
            hours_studied=row['Hours Studied'],
            previous_scores=row['Previous Scores'],
            extracurricular=row['Extracurricular Activities'] == 'Yes',
            sleep_hours=row['Sleep Hours'],
            sample_papers=row['Sample Question Papers Practiced'],
            performance_index=row['Performance Index'],
        )
```

Run from Django shell:

```
python manage.py shell
>>> from performance.load_data import run
>>> run()
```

4. Train & Save ML Model

Create `performance/train_model.py`

```
import pandas as pd
import joblib
from sklearn.model_selection import train_test_split
from sklearn.linear_modelimport LinearRegression
from sklearn.preprocessing import LabelEncoder


def train():
    df = pd.read_csv('dataset.csv')

    df['Extracurricular Activities'] = LabelEncoder().fit_transform(
        df['Extracurricular Activities']
    )

    X = df.drop('Performance Index', axis=1)
    y = df['Performance Index']

    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42
    )

    model = RandomForestRegressor(n_estimators=200, random_state=42)
    model.fit(X_train, y_train)

    joblib.dump(model, 'performance/model.pkl')
    print("Model trained and saved")
```

Run:

```
python manage.py shell
>>> from performance.train_model import train
>>> train()
```

5. Serializer (`performance/serializers.py`)

```python
from rest_framework import serializers
from .models import StudentPerformance

class StudentPerformanceSerializer(serializers.ModelSerializer):
    class Meta:
        model = StudentPerformance
        fields = '__all__'
```

6. Prediction API View

performance/views.py

```
import joblib
import numpy as np
from rest_framework.decorators import api_view
from rest_framework.response import Response

model = joblib.load('performance/model.pkl')

@api_view(['POST'])
def predict_performance(request):
    data = request.data

    features = np.array([[
        data['hours_studied'],
        data['previous_scores'],
        1 if data['extracurricular'] else 0,
        data['sleep_hours'],
        data['sample_papers'],
    ]])

    prediction = model.predict(features)[0]

    return Response({
        'predicted_performance_index': round(float(prediction),
2)
    })
```

7. URLs

performance/urls.py

```
from django.urls import path
from .views import predict_performance

urlpatterns = [
    path('predict/', predict_performance),
]
```

student_ml/urls.py

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path('api/', include('performance.urls')),
]
```

8. Example API Request

POST `/api/predict/`

```
{
  "hours_studied": 6,
  "previous_scores": 78,
  "extracurricular": true,
  "sleep_hours": 7,
  "sample_papers": 3
}
```

Response

```
{
  "predicted_performance_index": 72.45
}
```

9. Deployment Notes

- Store `model.pkl` outside repo or in `/performance/`
- Switch DB to **PostgreSQL**

10. Question

a. What is the purpose of joblib?
b. Mention other libraries that can achieve the same as joblib